Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

HHS Public Access

Author manuscript

Gut. Author manuscript; available in PMC 2025 November 11.

Published in final edited form as:

Gut. ; 73(12): 1953–1964. doi:10.1136/gutjnl-2024-332475.

Age-related patterns of microbial dysbiosis in multiplex inflammatory bowel disease families

Jonathan P. Jacobs[1,2,3], **Elizabeth A. Spencer**[4], **Drew Helmus**[5], **Julianne C. Yang**[1], **Venu Lagishetty**[1], **Gerold Bongers**[6,7], **Graham J. Britton**[6,7], **Kyle Gettler**[8], **Pamela Reyes-Mercedes**[5], **Jianzhong Hu**[6], **Amy Hart**[7], **Esi Lamousé-Smith**[7], **Jan Wehkamp**[7], **Carol Landers**[8], **Philip Debbas**[8], **Joana Torres**[5,9], **Jean-Frederic Colombel**[5], **Judy Cho**[10], **Inga Peter**[6], **Jeremiah Faith**[6,11], **Jonathan Braun**[8], **Marla C. Dubinsky**[4]

[1]Vatche and Tamar Manoukian Division of Digestive Diseases, Department of Medicine, David Geffen School of Medicine at UCLA, Los Angeles, California, USA

[2]Division of Gastroenterology, Hepatology and Parenteral Nutrition, VA Greater Los Angeles Healthcare System, Los Angeles, California, USA

[3]Goodman-Luskin Microbiome Center, David Geffen School of Medicine at UCLA, Los Angeles, California, USA

[4]The Division of Pediatric Gastroenterology and Nutrition, Icahn School of Medicine at Mount Sinai, New York, New York, USA

[5]The Dr. Henry D. Janowitz Division of Gastroenterology, Icahn School of Medicine at Mount Sinai, New York, New York, USA

Correspondence: JJacobs@mednet.ucla.edu, 10833 Le Conte Ave., CHS 42-210, Los Angeles, CA 90095.

Author Contributions

Jonathan P. Jacobs is the guarantor.

Jonathan P. Jacobs – Conceptualization, Methodology, Software, Validation, Formal Analysis, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization
Elizabeth A. Spencer – Conceptualization, Investigation, Data Curation, Writing – Review & Editing
Drew Helmus – Investigation, Data Curation, Writing – Review & Editing, Project Administration
Julianne Yang – Software, Formal Analysis
Venu Lagishetty – Formal Analysis, Investigation
Gerold Bongers – Investigation, Data Curation
Graham J. Britton – Investigation
Kyle Gettler – Investigation, Formal Analysis, Data Curation
Pamela Reyes-Mercedes – Investigation
Jianzhong Hu – Validation, Data Curation
Amy Hart – Conceptualization, Writing – Review & Editing
Esi Lamousé-Smith – Conceptualization, Writing – Review & Editing
Jan Wehkamp – Conceptualization
Carol Landers – Investigation
Philip Debbas – Investigation
Joana Torres – Investigation
Jean-Frederic Colombel – Conceptualization, Writing – Review & Editing
Judy Cho – Conceptualization, Writing – Review & Editing
Inga Peter – Validation, Writing – Review & Editing
Jeremiah Faith – Conceptualization, Writing – Review & Editing
Jonathan Braun – Conceptualization, Writing – Review & Editing, Supervision, Funding Acquisition
Marla C. Dubinsky – Conceptualization, Writing – Review & Editing, Supervision, Funding Acquisition

Competing Interests

Gerold Bongers, Amy Hart, Esi Lamousé-Smith, and Jan Wehkamp are current employees of Johnson & Johnson Innovative Medicine. Marla Dubinsky and Jean-Frederic Colombel are consultants for Johnson & Johnson Innovative Medicine and Prometheus Labs. All other authors do not have disclosures.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

[6]Department of Genetics and Genomic Sciences, Icahn School of Medicine at Mount Sinai, New York, New York, USA

[7]Marc and Jennifer Lipschultz Precision Immunology Institute, Icahn School of Medicine at Mount Sinai, New York, New York, USA

[8]Department of Pathology, Icahn School of Medicine at Mount Sinai, New York, New York, USA

[9]Janssen, Spring House, Pennsylvania, USA

[10]F. Widjaja Inflammatory Bowel Disease Institute, Cedars Sinai Medical Center, Los Angeles, California, USA

[11]Hospital da Luz, Lisboa, Portugal

Abstract

Objective: Inflammatory bowel disease (IBD) is characterized by dysbiosis, but it remains unclear to what extent dysbiosis develops in unaffected at-risk individuals. To address this, we investigated age-related patterns of fecal and serum markers of dysbiosis in high-risk multiplex IBD families (2 or more affected first-degree relatives).

Design: Fecal and serum samples were collected from multiplex IBD and control families (95 IBD, 292 unaffected, 51 controls). Findings were validated in independent cohorts of 616 and 1,173 subjects including patients with IBD, infants born to mothers with IBD, and controls. 16S rRNA gene sequencing and global untargeted metabolomics profiling of feces and serum were performed.

Results: Microbial and metabolomic parameters of dysbiosis progressively decreased from infancy until age 8. This microbial maturation process was slower in infants born to mothers with IBD. After age 15, dysbiosis steadily increased in unaffected relatives throughout adulthood. Dysbiosis was accompanied by marked shifts in the fecal metabolome and, to a lesser extent, the serum metabolome. Fecal and serum metabolomics dysbiosis indices validated in an independent cohort. Dysbiosis was associated with elevated anti-microbial serologies but not with fecal calprotectin. Dysbiosis metrics differentiated IBD from non-IBD comparably to serologies, with a model combining calprotectin, fecal metabolomics dysbiosis index, and serology score demonstrating highest accuracy.

Conclusion: These findings support that dysbiosis exists as a pre-disease state detectable by fecal and serum biomarkers for IBD risk prediction. Given expansion of disease-modifying agents and non-invasive imaging, the indices developed here may facilitate earlier diagnoses and improved management in at-risk individuals.

Keywords

Dysbiosis; microbial maturation; metabolomics; pre-disease biomarkers; risk prediction

INTRODUCTION

Inflammatory bowel disease (IBD) has been rising in incidence throughout the world and is now estimated to affect 1.3% of the adult population in the United States.[1] This has been

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

attributed to multiple environmental risk factors including diet, urban living, and antibiotic usage interacting with underlying genetic risk from over 200 IBD-associated loci.[2] It is widely believed that the intestinal microbiome is a critical mediator of the relationship of environmental and genetic risk factors with disease induction. This has been supported by the preponderance of IBD-associated genes involved in host-microbiome interactions and extensive preclinical research demonstrating that the microbiota can alter susceptibility to experimental models of IBD.[3] Prevailing models of IBD pathogenesis posit that there is a microbial risk state – often referred to as "dysbiosis" – that precedes the onset of intestinal inflammation and predisposes to its development due to a combination of reduced anti-inflammatory homeostatic signals and pathobiont potential within the microbial community.[4] Human microbiome association studies have consistently shown alterations in microbial composition and function in IBD patients, including at the time of disease diagnosis and during endoscopic remission.[5 6] However, it has been unclear to what extent observed microbial profiles in clinically diagnosed IBD patients are a cause of disease rather than representing a consequence of prior or ongoing intestinal inflammation.

To investigate whether dysbiosis occurs in those at high risk for IBD, studies have characterized the microbiome of unaffected first-degree relatives (FDRs). In an early study, we studied microbial composition of 21 pediatric IBD patients and their 69 FDRs.[7] We found by modeling microbial community structure that the microbiome of individuals in this cohort could be classified into two community states, termed OTU-types. The OTU-type with lower microbial diversity – consistent with a state of dysbiosis – was strongly associated with IBD but was also found in some unaffected FDRs, raising the possibility that it represented a pre-disease state. Recent data from the GEM study of unaffected FDRs of patients with Crohn's disease (CD) support that microbial changes can precede CD onset.[8] A microbiome risk score generated by machine learning from genus-level composition in samples obtained within approximately 3 years prior to diagnosis was associated with increased risk of developing CD. These findings are consistent with the hypothesis that pre-disease microbial changes promote risk for subsequent disease development, though it remains possible that the reported signature taxa were markers of a subclinical early disease phase.

To further define putative pre-disease dysbiosis in those at high risk for IBD, we performed a multi-omics analysis of microbial composition and function – including both fecal and serum global untargeted metabolomics – in a large cohort of Ashkenazi Jewish families with a history of multiplex IBD. We also assessed predictors of future IBD risk including anti-microbial serologic biomarkers and intestinal inflammation as assessed by fecal calprotectin.[8 9] Building on the previously reported microbial community states defined as OTU-types, we characterized an IBD-associated dysbiosis that had parallels with early life microbial maturation, appeared in unaffected family members during early adulthood, was associated with elevated anti-microbial serologies, and could be accurately predicted by fecal and serum metabolite markers.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

MATERIALS AND METHODS

Ashkenazi Jewish families with a history of multiplex IBD (2 or more affected first-degree relatives, including twins) and control (no known family history of IBD) Ashkenazi Jewish families were recruited from the greater New York metropolitan area by a combination of outreach to community leaders, posted fliers, and a series of educational and recruitment events in the communities. Stool was collected at home and sent by overnight shipping on ice packs to the research team. Blood samples were drawn from participants over age 7, and buccal swabs were obtained for those under 7. Longitudinal monitoring is ongoing to detect incident IBD cases. The study was approved by the Mount Sinai IRB (STUDY-16–00021). Validation was performed using the previously described MECONIUM and MSCCR cohorts.[10 11] Fecal 16S rRNA gene sequencing, fecal calprotectin, fecal and serum global untargeted metabolomics, serologies, and polygenic risk score along with bioinformatics analysis are described in the Supplementary Materials. Community members were involved in study design and conduct. Research questions reflected their interest in understanding factors that increase their risk of IBD. Study protocols were optimized with community input to minimize participant burden.

RESULTS

IBD-associated dysbiosis occurs among unaffected members of multiplex IBD families

Road to Prevention (RTP) is a family-based cohort consisting of multiplex IBD and control Ashkenazi Jewish families. Stool samples were obtained from 387 individuals from 45 multiplex IBD families – including 64 with CD, 10 with ulcerative colitis (UC), 21 with indeterminate colitis (IBDU), and 292 unaffected family members – and 51 individuals from 10 control families (Tables 1, S1). Fecal calprotectin was elevated in CD and UC/IBDU compared to unaffected family members ($p=7\times10^{-10}$, p=0.0005), which did not differ from controls (Table 1, Fig. S1). The intestinal microbiome was assessed by 16S rRNA gene sequencing. Patients with CD and UC/IBDU had significantly lower microbial diversity by measures of richness ($p=7\times10^{-7}$, p=0.0006) and evenness ($p=4\times10^{-8}$, $p=6\times10^{-5}$) compared to unaffected family members, which were comparable to controls (p=0.64, p=0.31) (Fig. 1A, S2A). Beta diversity analysis demonstrated significant association of microbial composition with IBD group (CD, UC/IBDU, unaffected, controls) in those over 3 years of age (Fig. 1B–C, S3A).

The distribution of subjects visualized by principal coordinates analysis paralleled that of our earlier family cohort study, in which fecal microbiota were classified into two microbial community states termed OTU-types (Fig. 1B).[7] A random forest classifier trained on the prior study was used to define OTU-types in RTP (Fig. S4A). Subjects with OTU-type 1 were clearly separated from those with OTU-type 2, which was enriched in CD (59%) and UC/IBDU (56%) (Fig. 1B, 1D). OTU-type 2 showed significantly lower microbial richness and evenness compared to OTU-type 1 within each group while there was no significant difference among groups within OTU-type 1, matching our prior findings (Fig. 1A, S2B).[7] As in the initial study, there was a subset of unaffected family members with dysbiosis as defined by the IBD-associated OTU-type 2.[7] This was more prevalent than in the controls (13% vs. 5% above age 3), though due to the comparatively small

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

number of controls this difference did not reach significance (p=0.17) (Fig. 1D). There was no difference in fecal calprotectin between OTU-types within each group, supporting that it was not an indicator of microbial changes induced by intestinal inflammation (Fig. S5A). OTU-type was also not associated with polygenic risk score for IBD, suggesting that it was not primarily a genetically encoded state (Fig. S5B). Beta diversity analysis demonstrated a highly significant association of microbial composition with OTU-type but loss of significant association with IBD after accounting for dysbiosis as represented by OTU-types (Fig. 1C). Similarly, differential abundance testing demonstrated only a limited number of taxa associated with CD (14 ASVs) or UC/IBDU (6 ASVs) after adjusting for OTU-type (Fig. 1E, S6). In contrast, OTU-type 2 was associated with shifts in 168 ASVs, including expansion of *Escherichia-Shigella*, *Ruminococcus gnavus*, *Actinomyces*, *Streptococcus*, and *Enterococcus* ASVs and depletion of *Faecalibacterium*, *Roseburia*, *Coprococcus, Bacteroides*, *Akkermansia*, *Bifidobacterium*, and *Collinsella* ASVs (Fig. 1F).

Dysbiosis in unaffected family members arises in late adolescence and parallels early life maturation, which is delayed in infants born to IBD mothers

Assessing OTU-type prevalence by age in unaffected family members, we observed a striking pattern in which the health-associated OTU-type 1 was absent from infants below 1 year of age but became increasingly prevalent over subsequent age strata, reaching a peak of 100% prevalence between 8 and 15 years of age (Fig. 2A). Starting at 15–20 years of age (adolescence to early adulthood transition), there was a progressive decrease in OTU-type 1 prevalence with increasing age among unaffected family members. Decreased prevalence of OTU-type 1 was also seen among the IBD patients in age strata over age 15 compared to younger IBD patients. To quantify degree of dysbiosis, we developed a metric based upon Bray-Curtis dissimilarity that captured differences between OTU-types 1 and 2 (Fig. S7). This dysbiosis score was highest in the first year of life, rapidly decreased to a trough by age 8, then steadily increased starting around age 20 among unaffected family members (Fig. 2B). Considering all ages above 3, dysbiosis score was non-significantly increased in unaffected family members compared to controls (p=0.17) and showed a highly significant positive association with age in unaffected family members ($p=7\times10^{-9}$) (Fig. 2C).

The relationship of OTU-type and dysbiosis score with age during the first several years of life suggested that they captured early life microbial maturation (Fig. S3). However, RTP lacked longitudinal sampling during early life and contained too few control infants to determine if maturation differed in infants born to families with a history of IBD. To assess this, we applied OTU-types and dysbiosis score to the longitudinal MECONIUM cohort of 105 infants born to mothers with IBD and 248 control infants (Table S2). OTU-type 1 was absent after birth but showed increasing prevalence starting at month 2 which was higher in infants of control families than infants of mothers with IBD. This difference persisted through 4 years of age, with highly significant difference between the two groups across all time points (p=0.0015) (Fig. 2D). Similarly, both groups of infants showed high dysbiosis scores shortly after birth that steadily decreased during the first four years of life, but this reduction occurred faster in control infants with separation of the groups by 18 months of age (p=0.0036 across all time points) (Fig. 2E). The association of maternal IBD with increased dysbiosis score in infants remained significant after adjusting for covariates

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

affecting dysbiosis score including delivery mode, microbial diversity, and abundance of *Escherichia-Shigella* (Fig. S8).

To validate the relationship of age with increased dysbiosis score throughout adulthood, we then applied OTU-types and dysbiosis score to the parents and other family members in the MECONIUM cohort as well as the MSCCR cohort of IBD patients and community controls (Fig. S5B, S9–S10). Fecal 16S rRNA sequence data were available for 226 non-IBD, 16 CD, and 21 UC individuals in MECONIUM and 150 CD, 117 UC/IBDU, and 72 controls in MSCCR (Table S2–S3). The MECONIUM and MSCCR cohorts contained complementary age distributions and were combined to span the spectrum of adult ages. When considered together, these two cohorts validated the finding of progressively increasing dysbiosis throughout adulthood ($p=7\times10^{-5}$ for age adjusting for cohort and sex) (Fig. 2F).

Dysbiosis has characteristic fecal and serum metabolite signatures

We assessed the functional consequences of intestinal dysbiosis by performing global untargeted metabolomics analysis of fecal samples. Metabolomics profiles were significantly associated with IBD status and partially separated IBD patients from unaffected family members and controls over age 3, mirroring the pattern seen with microbiome composition (Fig. 3A–B, S11). Given the similarity between the fecal metabolome and microbiome, the two datasets were directly compared by Procrustes superimposition and found to be highly correlated (Mantel $r=0.67$, $p<10^{-5}$) (Fig. 3C). This supported that microbial community structure as represented by OTU-types could be a driver of metabolomic differences. Indeed, metabolomics profiles were more strongly associated with OTU-type than IBD group or fecal calprotectin (Fig. 3B). Short chain fatty acids (SCFAs) were then analyzed and found to be depleted in IBD, with significant reduction in all eight measured SCFAs in UC/IBDU and in hexanoate and valerate in CD (Fig. 3D, Fig. S12A–B). OTU-type was associated with significant reductions in butyrate, isobutyrate, isovalerate, 2-methylbutyrate, valerate, and hexanoate after adjusting for IBD status. Multivariate analysis of other metabolites demonstrated significant shifts in 680 metabolites between OTU-types spanning a wide range of pathways in amino acid, carbohydrate, lipid, nucleotide, and xenobiotic metabolism (Fig. 3E). CD but not UC/IBDU had a large metabolomics signature with 331 differential metabolites relative to unaffected relatives, of which 207 overlapped with OTU-type 2 (Fig. S13). This suggests that CD accentuates the metabolite alterations related to dysbiosis.

Given the strong association of OTU-types with the fecal metabolome, we constructed a random forest classifier which achieved high accuracy to predict OTU-type from fecal metabolites (AUC 0.97 overall, AUC 0.91 for unaffected) (Fig. 3F, Fig. S14A). It included 93 metabolites spanning a wide range of metabolic pathways including fatty acid, amino acid, nucleotide, and bile acid metabolism (Fig. S15). A dysbiosis index was then created, which was calculated as the ratio of metabolites in the classifier that were enriched in OTU-type 2 to those that were depleted in OTU-type 2. This fecal metabolomics dysbiosis index strongly differentiated CD and UC/IBDU from unaffected family members across all ages ($p<2\times10^{-16}$, $p=5\times10^{-7}$) and increased with age in unaffected family members throughout adulthood ($p=4\times10^{-5}$) as had been seen with dysbiosis score (Fig. 3G). Unaffected family

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

members had non-significantly higher fecal metabolomics dysbiosis index compared to controls (p=0.17). The fecal metabolite signature of dysbiosis was validated in MSCCR. The random forest classifier accurately distinguished OTU-types (AUC 0.90) and the fecal metabolomics dysbiosis index demonstrated similar elevation in CD and UC compared to controls as was observed in RTP ($p=3\times10^{-15}$, $p=2\times10^{-7}$) (Fig. 3H–I, Fig. S14B).

We then characterized serum metabolomics profiles due to the potential for microbial metabolites to be absorbed into systemic circulation. Separation was observed among IBD groups and between OTU-types, with OTU-type showing greater statistical significance ($p<10^{-5}$ vs. p=0.021) (Fig. 4A–B). This association with dysbiosis reflected the overall correlation of the serum metabolome with the fecal microbiome and metabolome (Mantel r=0.31 and r=0.35; both $p<10^{-5}$) (Fig. 4C). Differential abundance testing demonstrated significant changes in 42 serum metabolites by OTU-type, the majority belonging to pathways related to microbial metabolism such as secondary bile acid, tryptophan, and benzoate metabolism (Fig. 4D, S16A). In contrast, CD had only 3 significant metabolites and UC only 1 (Fig. S16). Serum short chain fatty acids were not significantly associated with IBD status or OTU-type (Fig. S12C–E). A random forest classifier predicted OTU-types from serum metabolites with AUC 0.94 (AUC 0.91 for unaffected) (Fig. 4E, Fig. S14C). Strikingly, the majority of the 22 metabolites in the classifier are known to be affected by microbial metabolism including the top 4 metabolites by importance score: hydrocinnamate, isoursodeoxycholate, p-cresol sulfate, and hippurate (Fig. 4G). Seven out of twelve metabolites that were detectable in feces showed OTU-type associations in the same direction as in serum (Fig. S17). The classifier metabolites were then used to generate a serum metabolomics dysbiosis index, which was significantly elevated in CD and UC/IBDU compared to unaffected relatives ($p<2\times10^{-16}$, $p=2\times10^{-6}$) (Fig. 4F). Serum metabolomics dysbiosis index trended higher in unaffected family members compared to controls (p=0.09) and was not significantly associated with age (p=0.39). The serum metabolite signature of dysbiosis was validated using serum metabolomics data from 470 CD, 374 UC, and 329 control subjects in MSCCR (Table S4). The random forest classifier demonstrated moderate accuracy to predict OTU-types with AUC 0.72 (Fig. 4H, Fig. S14D). Serum metabolomics dysbiosis index was significantly increased in CD compared to controls across ages ($p=8\times10^{-12}$), but there was no difference between UC/IBDU and controls (Fig. 4I).

Dysbiosis is associated with increased anti-microbial serologies

It has previously been reported that increased levels of anti-microbial serologies predict future risk of IBD.[9] To assess for a relationship of dysbiosis with serologic responses, we measured levels of Cbir1 (anti-flagellin), OmpC (outer membrane protein of *E. coli*), IgA and IgG ASCA (anti-*Saccharomyces*), and ANCA. All five serologies were significantly higher in CD and UC/IBDU than unaffected relatives and controls, which did not significantly differ (Fig. S18A). Principal components analysis was performed to aggregate the five serologies and identify global patterns that were not apparent from individual markers (Fig. 5A). Serology score was defined as the first principal component, which separated IBD from unaffected relatives and incorporated all five markers with greatest contribution from IgA and IgG ASCA (Fig. 5B). It was significantly higher in CD compared

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

to UC/IBDU, which in turn was significantly higher than unaffected relatives and controls; there was no difference between unaffected relatives and controls (Fig. S18B). Increased serology score was observed in unaffected relatives with OTU-type 2 compared to OTU-type 1 (Fig. 5C). Consistent with this, dysbiosis score and fecal metabolomics dysbiosis index were significantly associated with increased serology score (p=0.008, p=0.047), though there was no significant association with serum metabolomics dysbiosis index (Fig. 5D–F). Only one microbe (a *Subdoligranulum* ASV depleted in OTU-type 2) and no fecal or serum metabolites were associated with serology score in multivariate models adjusting for OTU-type (Fig. S18C).

Models incorporating fecal and serum dysbiosis metrics can estimate IBD risk

As IBD-associated dysbiosis may signify a pre-disease state, dysbiosis metrics have the potential to risk stratify unaffected members of multiplex IBD families. Logistic regression models were created to differentiate IBD from unaffected family members based upon fecal calprotectin, serology score, and dysbiosis metrics. Fecal calprotectin had relatively poor discriminative ability with AUC 0.67 whereas serology score was more robust with AUC 0.86. Dysbiosis score and fecal metabolomics dysbiosis index had comparable accuracy as serology score with AUC 0.84 and 0.86, while serum metabolomics dysbiosis index had somewhat less accuracy with AUC 0.76 (Fig. 6A). These five markers were then combined in three distinct risk models that incorporated markers from feces, serum, or both. A combined fecal classifier with fecal calprotectin, dysbiosis score, and fecal metabolomics dysbiosis index achieved AUC 0.87 with fecal metabolomics dysbiosis index contributing the most to the regression model and no significant additional contribution of dysbiosis score (Fig. 6B). A serum classifier incorporating serology score and serum metabolomics dysbiosis index achieved numerically higher accuracy with AUC 0.90 and significant contributions of both components (Fig. 6C). The greatest accuracy for differentiating IBD from non-IBD (AUC 0.96) was achieved with a combined fecal and serum classifier incorporating fecal calprotectin, fecal metabolomics dysbiosis index, and serology score (Fig. 6D). In all three models, there were unaffected family members with high (>50%) or intermediate (25–50%) model-predicted probability of IBD, potentially suggesting a pre-disease risk state or subclinical disease (Table S5). We examined risk predictions for the initial three incident IBD cases among unaffected family members in RTP (Fig. 6E). Two out of three demonstrated >50% predicted IBD risk by the fecal IBD risk model incorporating dysbiosis metrics. Of these, one also had serum data available, with serum and combined serum and fecal models showing >50% predicted risk.

DISCUSSION

We defined a multi-omics compositional and functional signature of dysbiosis in multiplex IBD families. Dysbiosis emerged in this cohort in late adolescence to early adulthood, overlapping with the median age of IBD onset in our cohort (19 years of age) and the reported age range of peak IBD incidence.[12] The majority of IBD patients had dysbiosis, but its onset in unaffected members of some multiplex IBD families likely did not reflect a subclinical mucosal inflammatory process as there was no relationship with fecal calprotectin. Among unaffected family members, dysbiosis was associated with elevated

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

levels of serologic markers of immune response against microbial antigens, a known pre-disease indicator.[9] However, dysbiosis may not be a driver of serologic profiles in IBD by itself, as a study of circulating antibody repertoires of IBD patients did not find any associations with fecal microbiota.[13] Instead, dysbiosis within multiplex IBD families likely signifies increased likelihood not just of IBD but also its pre-disease stages including development of anti-microbial serologies. Supporting the potential utility of dysbiosis as an indicator of a pre-IBD susceptibility state, models incorporating dysbiosis metrics predicted high risk in two out of three initial incident IBD cases.

It is unknown what could explain the onset of dysbiosis in late adolescence, as microbial changes during adolescence have received little study compared to those in early life. Longitudinal changes in microbiome have been reported following puberty; however, dysbiosis was not detected in the family cohort until after age 15, past the typical age of pubertal onset.[14] Development of dysbiosis may reflect changes that occur during the late adolescence to young adulthood transition, including dietary alteration, initiation of smoking, altered sleep patterns, increased stress, chronic antibiotic exposure for conditions such as acne, and other lifestyle shifts as individuals gain greater autonomy and leave their family homes.[15] In support of this, a longitudinal study of students during their freshman year of college demonstrated significant shifts in microbial composition.[16]

We further found that dysbiosis became more prevalent with increasing age. This may reflect the cumulative effect of environmental exposures over time resulting in progressive dysbiosis throughout adulthood. This does not appear related to underlying genetic risk given the lack of association with IBD-related variants summarized by polygenic risk score. Control Ashkenazi families in RTP as well as the adult controls in the two validation cohorts also showed increasing dysbiosis with age, suggesting that factors driving dysbiosis are common and that dysbiosis alone is insufficient to induce IBD. Prior studies have reported that age is a significant factor explaining variation in microbial composition across adult populations.[17] Interestingly, some of the taxa that are depleted with dysbiosis have also been reported to be consistently depleted with increased age, including *Faecalibacterium* and *Roseburia*.[18]

Surprisingly, there was a clear relationship of dysbiosis metrics with early life microbial maturation. Dysbiosis in later life may parallel a disordered microbiome in early life that has not matured to a stable adult-like state. Intriguingly, children of mothers with IBD were found to have delays in attainment of a healthy microbial community state as assessed by OTU-types as well as dysbiosis score. This delay in microbial maturation is consistent with earlier reports from MECONIUM that infants of mothers with IBD have lower diversity microbiota than controls.[10] It is unknown whether delayed microbial maturation confers increased likelihood of developing dysbiosis in early adulthood and may therefore contribute to IBD risk in offspring of mothers with IBD.[19] This would be an important area of future study as it raises the possibility that dysbiosis risk in adulthood could be modulated by microbiome interventions in early life.

Dysbiosis was detectable not just by microbial composition but also by robust shifts in the fecal metabolome. This included changes previously associated with IBD such as

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

enrichment of cholate and its conjugates, depletion of the secondary bile acid lithocholate, and enrichment of acyl carnitines.[20] Tryptophan and its metabolites were also notably associated with dysbiosis. Many microbial-derived tryptophan catabolites are agonists of the aryl hydrocarbon receptor (Ahr), which modulates mucosal immune function.[21] A pattern of increased kynurenine and decreased Ahr ligands has been reported in IBD and may promote vulnerability to intestinal inflammation.[22] Among the SCFAs, dysbiosis was associated specifically with reduction in butyrate but not acetate or propionate. Butyrate is known to have wide-ranging anti-inflammatory effects that reduce severity of experimental IBD models, supporting that dysbiosis favors increased inflammatory tone.[23] There was also prominent alteration in phenolic compounds, which arise from microbial metabolism and were recently found to also be altered in irritable bowel syndrome patients.[24] Serum metabolites similarly showed a highly significant relationship with dysbiosis, with concordant changes in the fecal and serum metabolome of known bacterial products – including phenolic compounds and bile acids – supporting that serum levels directly reflected intestinal microbial metabolism.

Strengths of this study included the unique risk profiles of multiplex IBD families, multi-omics approach including paired fecal and serum metabolomics, large sample size with two validation cohorts, and integration of dysbiosis measures into risk models with other known pre-disease markers. A current limitation is the small number of incident IBD cases; continued longitudinal follow-up is ongoing and increased numbers of incident cases in the future would allow further validation of the relationship of dysbiosis defined by microbial and metabolite markers to IBD risk. Another limitation is the comparatively small number of control families, reducing power to detect differences in dysbiosis between unaffected family members and controls. Moreover, the generalizability of findings from the RTP cohort to other at-risk IBD populations may be questioned; importantly, the fecal and serum dysbiosis metrics were applicable to the mixed ethnicity MECONIUM and MSCCR cohorts.

In conclusion, we demonstrated that dysbiosis is detectable in at-risk unaffected individuals from multiplex IBD families starting in late adolescence and becomes increasingly prevalent throughout adulthood. We further established the feasibility of both stool and blood-based microbial metabolite measurement to assess for dysbiosis in at-risk populations. This will require longitudinal validation but raises the prospect of prospectively identifying microbially at-risk individuals who may benefit from microbiota-targeted interventions and non-invasive imaging for early disease detection. Reproducible characterization of microbial community features that are consistent with dysbiosis will also promote preclinical studies to characterize specific microbes and their products within dysbiotic microbial communities that increase IBD susceptibility. This makes it feasible to identify prophylactic targets for IBD prevention as well as biomarkers of risk, bringing us further along down the road to prevention of IBD after decades of rising incidence worldwide.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Acknowledgements

We would like to thank all the families who participated in this study and the Jewish Crohn's and Colitis Support Group, led by Yoely Drummer.

Funding

This study was funded by a grant from Janssen Research and Development, LLC, and by a generous donation from Julie and Bruce Goldsmith. The MECONIUM study was supported by the Crohn's and Colitis Foundation. This work was also supported in part through the computational and data resources and staff expertise provided by Scientific Computing and Data at the Icahn School of Medicine at Mount Sinai and supported by the Clinical and Translational Science Awards (CTSA) grant UL1TR004419 from the National Center for Advancing Translational Sciences. JPJ was supported by VA IK2CX001717. MCD and JC were supported by R01 DK123758-01 from the National Institute of Diabetes and Digestive and Kidney Diseases.

Data Availability Statement

All data relevant to the study are included in the article or uploaded as supplementary information. Sequence data have been deposited in NCBI Bioproject under PRJNA1053656 (https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1053656) and metabolomics data along with associated metadata are included in supplemental data files.

REFERENCES

1. Dahlhamer JM, Zammitti EP, Ward BW, et al. Prevalence of Inflammatory Bowel Disease Among Adults Aged >/=18 Years - United States, 2015. MMWR Morb Mortal Wkly Rep 2016;65(42):1166–69. doi: 10.15585/mmwr.mm6542a3 [published Online First: 2016/10/28] [PubMed: 27787492]
2. Piovani D, Danese S, Peyrin-Biroulet L, et al. Environmental Risk Factors for Inflammatory Bowel Diseases: An Umbrella Review of Meta-analyses. Gastroenterology 2019;157(3):647–59 e4. doi: 10.1053/j.gastro.2019.04.016 [published Online First: 2019/04/25] [PubMed: 31014995]
3. Shan Y, Lee M, Chang EB. The Gut Microbiome and Inflammatory Bowel Diseases. Annu Rev Med 2022;73:455–68. doi: 10.1146/annurev-med-042320-021020 [published Online First: 2021/09/24] [PubMed: 34555295]
4. Sartor RB, Wu GD. Roles for Intestinal Bacteria, Viruses, and Fungi in Pathogenesis of Inflammatory Bowel Diseases and Therapeutic Approaches. Gastroenterology 2017;152(2):327–39 e4. doi: 10.1053/j.gastro.2016.10.012 [published Online First: 2016/10/23] [PubMed: 27769810]
5. Gevers D, Kugathasan S, Denson LA, et al. The treatment-naive microbiome in new-onset Crohn's disease. Cell Host Microbe 2014;15(3):382–92. doi: 10.1016/j.chom.2014.02.005 [published Online First: 2014/03/19] [PubMed: 24629344]
6. Jacobs JP, Goudarzi M, Lagishetty V, et al. Crohn's disease in endoscopic remission, obesity, and cases of high genetic risk demonstrates overlapping shifts in the colonic mucosal-luminal interface microbiome. Genome Med 2022;14(1):91. doi: 10.1186/s13073-022-01099-7 [published Online First: 2022/08/16] [PubMed: 35971134]
7. Jacobs JP, Goudarzi M, Singh N, et al. A Disease-Associated Microbial and Metabolomics State in Relatives of Pediatric Inflammatory Bowel Disease Patients. Cell Mol Gastroenterol Hepatol 2016;2(6):750–66. doi: 10.1016/j.jcmgh.2016.06.004 [published Online First: 2017/02/09] [PubMed: 28174747]
8. Raygoza Garay JA, Turpin W, Lee SH, et al. Gut Microbiome Composition Is Associated With Future Onset of Crohn's Disease in Healthy First-Degree Relatives. Gastroenterology 2023;165(3):670–81. doi: 10.1053/j.gastro.2023.05.032 [published Online First: 2023/06/02] [PubMed: 37263307]
9. Torres J, Petralia F, Sato T, et al. Serum Biomarkers Identify Patients Who Will Develop Inflammatory Bowel Diseases Up to 5 Years Before Diagnosis. Gastroenterology 2020;159(1):96–104. doi: 10.1053/j.gastro.2020.03.007 [published Online First: 2020/03/14] [PubMed: 32165208]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

10. Torres J, Hu J, Seki A, et al. Infants born to mothers with IBD present with altered gut microbiome that transfers abnormalities of the adaptive immune system to germ-free mice. Gut 2020;69(1):42–51. doi: 10.1136/gutjnl-2018-317855 [published Online First: 2019/05/01] [PubMed: 31036757]
11. Argmann C, Hou R, Ungaro RC, et al. Biopsy and blood-based molecular biomarker of inflammation in IBD. Gut 2023;72(7):1271–87. doi: 10.1136/gutjnl-2021-326451 [published Online First: 2022/09/16] [PubMed: 36109152]
12. Johnston RD, Logan RF. What is the peak age for onset of IBD? Inflamm Bowel Dis 2008;14 Suppl 2:S4–5. doi: 10.1002/ibd.20545 [published Online First: 2008/09/26] [PubMed: 18816745]
13. Bourgonje AR, Andreu-Sanchez S, Vogl T, et al. Phage-display immunoprecipitation sequencing of the antibody epitope repertoire in inflammatory bowel disease reveals distinct antibody signatures. Immunity 2023;56(6):1393–409 e6. doi: 10.1016/j.immuni.2023.04.017 [published Online First: 2023/05/11] [PubMed: 37164015]
14. Korpela K, Kallio S, Salonen A, et al. Gut microbiota develop towards an adult profile in a sex-specific manner during puberty. Sci Rep 2021;11(1):23297. doi: 10.1038/s41598-021-02375-z [published Online First: 2021/12/04] [PubMed: 34857814]
15. McVey Neufeld SF, Ahn M, Kunze WA, et al. Adolescence, the microbiota-gut-brain axis, and the emergence of psychiatric disorders. Biol Psychiatry 2023 doi: 10.1016/j.biopsych.2023.10.006 [published Online First: 2023/10/16]
16. Mohr AE, Ahern MM, Sears DD, et al. Gut microbiome diversity, variability, and latent community types compared with shifts in body weight during the freshman year of college in dormitory-housed adolescents. Gut Microbes 2023;15(2):2250482. doi: 10.1080/19490976.2023.2250482 [published Online First: 2023/08/29] [PubMed: 37642346]
17. Zhernakova A, Kurilshikov A, Bonder MJ, et al. Population-based metagenomics analysis reveals markers for gut microbiome composition and diversity. Science 2016;352(6285):565–9. doi: 10.1126/science.aad3369 [published Online First: 2016/04/30] [PubMed: 27126040]
18. Ghosh TS, Shanahan F, O'Toole PW. The gut microbiome as a modulator of healthy ageing. Nat Rev Gastroenterol Hepatol 2022;19(9):565–84. doi: 10.1038/s41575-022-00605-x [published Online First: 2022/04/27] [PubMed: 35468952]
19. Dotan I, Alper A, Rachmilewitz D, et al. Maternal inflammatory bowel disease has short and long-term effects on the health of their offspring: a multicenter study in Israel. J Crohns Colitis 2013;7(7):542–50. doi: 10.1016/j.crohns.2012.08.012 [published Online First: 2012/10/06] [PubMed: 23036507]
20. Lloyd-Price J, Arze C, Ananthakrishnan AN, et al. Multi-omics of the gut microbial ecosystem in inflammatory bowel diseases. Nature 2019;569(7758):655–62. doi: 10.1038/s41586-019-1237-9 [published Online First: 2019/05/31] [PubMed: 31142855]
21. Roager HM, Licht TR. Microbial tryptophan catabolites in health and disease. Nat Commun 2018;9(1):3294. doi: 10.1038/s41467-018-05470-4 [published Online First: 2018/08/19] [PubMed: 30120222]
22. Lamas B, Richard ML, Leducq V, et al. CARD9 impacts colitis by altering gut microbiota metabolism of tryptophan into aryl hydrocarbon receptor ligands. Nat Med 2016;22(6):598–605. doi: 10.1038/nm.4102 [published Online First: 2016/05/10] [PubMed: 27158904]
23. Deleu S, Machiels K, Raes J, et al. Short chain fatty acids and its producing organisms: An overlooked therapy for IBD? EBioMedicine 2021;66:103293. doi: 10.1016/j.ebiom.2021.103293 [published Online First: 2021/04/05] [PubMed: 33813134]
24. Jacobs JP, Lagishetty V, Hauer MC, et al. Multi-omics profiles of the intestinal microbiome in irritable bowel syndrome and its bowel habit subtypes. Microbiome 2023;11(1):5. doi: 10.1186/s40168-022-01450-5 [published Online First: 2023/01/10] [PubMed: 36624530]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

What is already known on this topic:

- While dysbiosis is accepted as a notable feature of IBD, it's unclear to what extent IBD-associated dysbiosis occurs in the absence of disease in those at risk.

What this study adds:

- Dysbiosis was detected in unaffected individuals from multiplex IBD families starting in late adolescence and was associated with fecal and serum metabolite profiles.
- Early life maturation could be captured by dysbiosis metrics and was delayed in infants of mothers with IBD.
- Dysbiosis in unaffected members of multiplex IBD families was associated with elevated anti-microbial serologies and could be incorporated into fecal or serum biomarker risk models for IBD.

How this study might affect research, practice or policy:

- Patterns of intestinal dysbiosis in unaffected members of multiplex IBD families suggest that it is insufficient to induce IBD but may represent a pre-disease susceptibility state that promotes disease onset in those with underlying risk.
- It may be feasible to prospectively identify microbially at-risk individuals who could benefit from microbiota-targeted interventions.

Author Manuscript
Author Manuscript
Author Manuscript
Author Manuscript

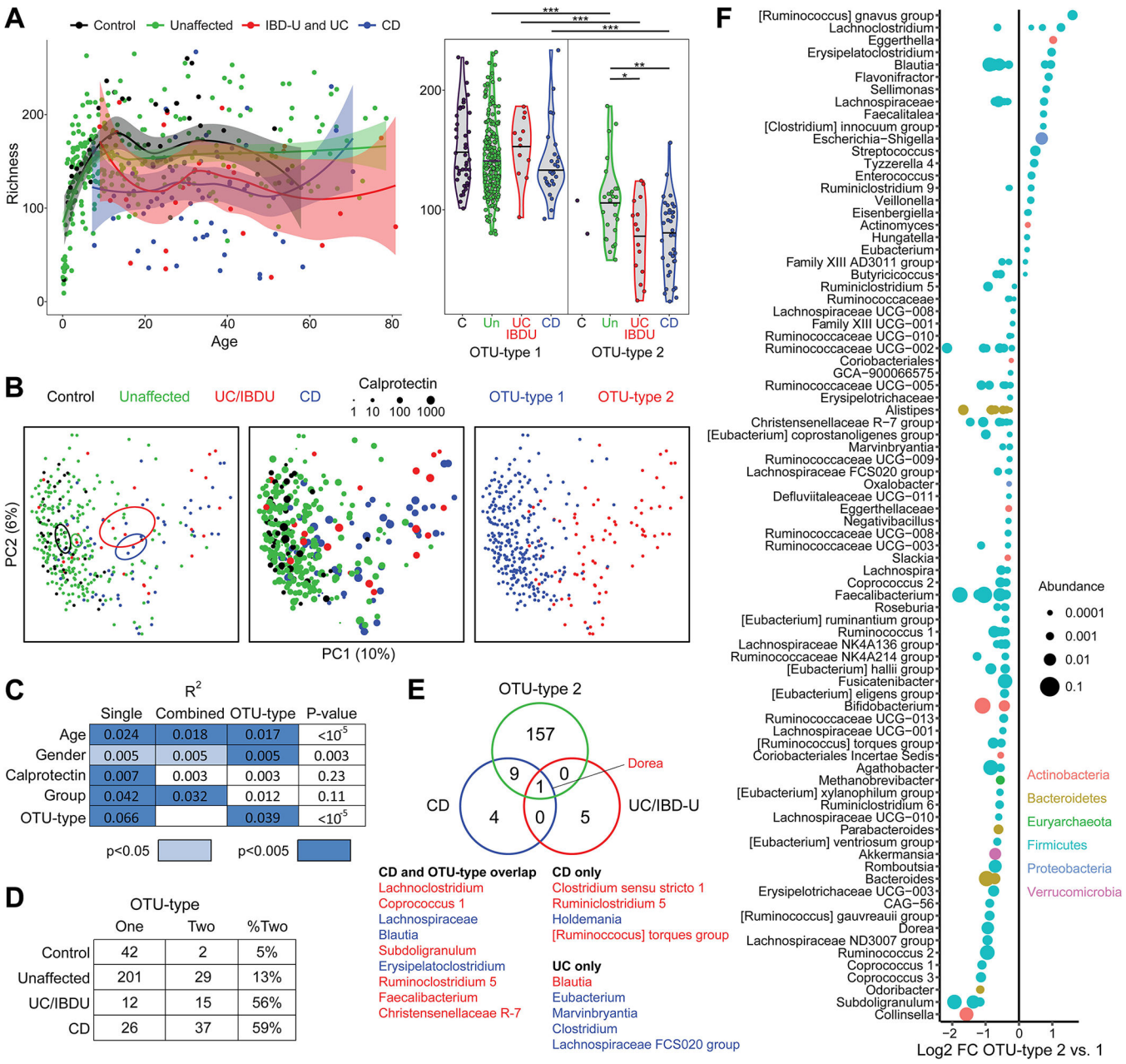


Figure 1. Unaffected members of multiplex IBD families can carry an IBD-associated microbial community state representing dysbiosis (OTU-type 2).

(**A**) Microbial richness (number of ASVs) by age is shown for controls (C), unaffected family members (Un), ulcerative colitis and unclassified IBD (UC/IBD-U), and Crohn's disease (CD). Lines were fitted by loess; shaded areas represent the 95% confidence interval. Richness of subjects greater than 3 years of age is shown by violin plots, divided by group and OTU-type. *p<0.05, **p< 0.01, ***p<0.0001 (**B**) Principal coordinate analysis (PCoA) plots visualizing differences in microbial composition measured by Bray-Curtis dissimilarity. Each dot represents one subject, colored by IBD group (left and middle panels) or OTU-type (right panel). Ellipses indicate the border of the 95% confidence interval for IBD group centroids in the left panel. Dot size is proportional to fecal calprotectin in the

middle panel. Subjects under 3 years of age were excluded. (**C**) Effect size (R^2) of variables in univariate ("single") and multivariate PERMANOVA models ("combined" = all variables except OTU-type; "OTU-type" = all variables) for Bray-Curtis dissimilarity. The marginal R^2 and p-value are shown for multivariate models. Significance is indicated by color and p-values are shown for the model incorporating OTU-type. (**D**) Distribution of OTU-types within each IBD group. (**E**) Venn diagram of differentially abundant ASVs in OTU-type 2 vs. 1, UC/IBD-U vs. unaffected, and CD vs. unaffected. Genus names are shown for IBD-associated ASVs; blue text indicates enriched and red indicates depleted taxa. (**F**) Log2 fold change (FC) of differentially abundant ASVs between OTU-type 2 vs. 1 in subjects over age 3 adjusting for sex, age, IBD group, and family. ASVs are represented by dots and organized by genus.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

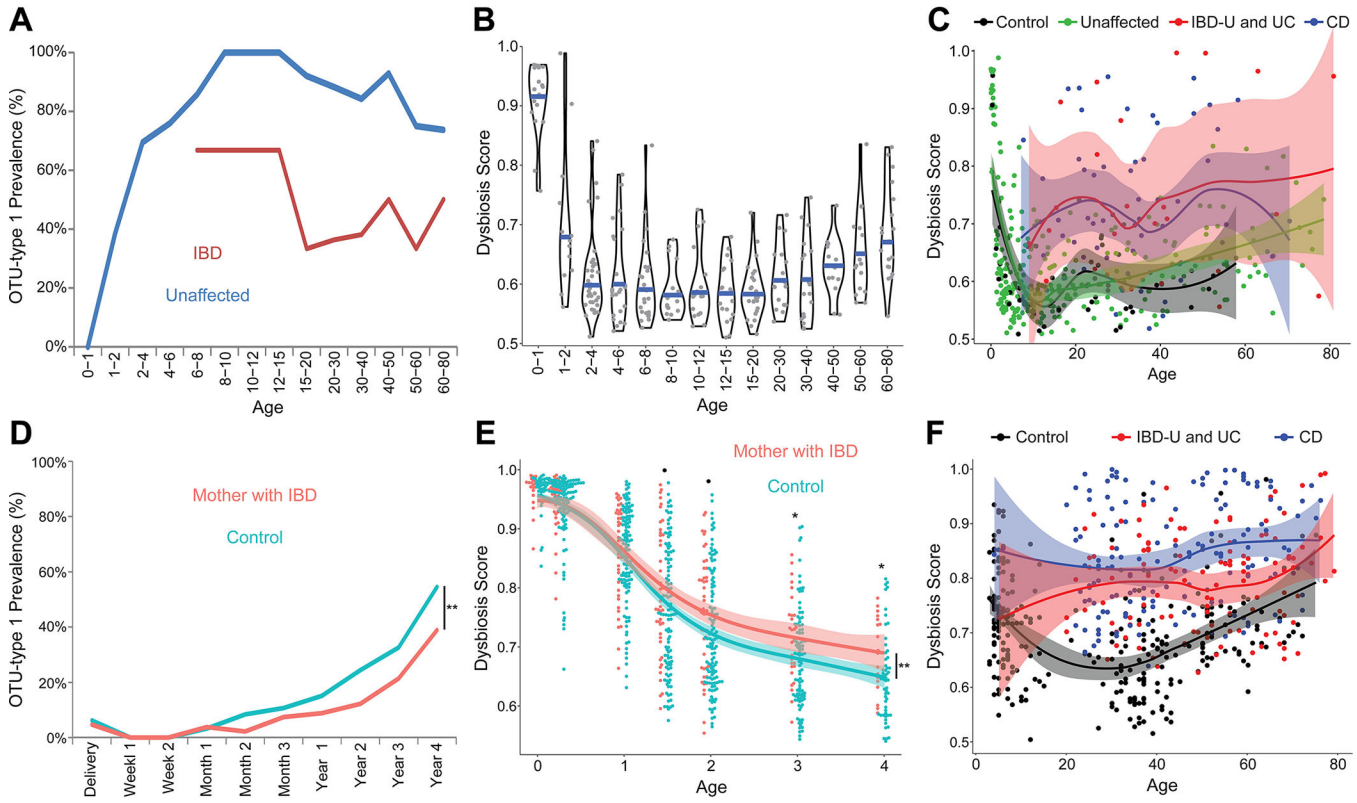


Figure 2. IBD families show slower microbiota maturation in early life and progressive dysbiosis starting in early adulthood.

(**A**) OTU-type 1 prevalence in IBD and unaffected family members by age strata. (**B**) Dysbiosis score (representing the median Bray-Curtis dissimilarity to controls with OTU-type 1) in unaffected family members by age strata. (**C**) Dysbiosis score by age and IBD group. (**D**) OTU-type 1 prevalence by age in the MECONIUM cohort of children born to mothers with IBD or controls. Significance across all time points determined by mixed effects logistic regression adjusting for sex and age. **p<0.01 (**E**) Dysbiosis score by age in the MECONIUM cohort. Significance across all time points assessed by a mixed effects model adjusting for sex and age; significance at individual time points assessed by linear models adjusting for sex. •p<0.1 *p<0.05 (**F**) Dysbiosis score by age in the MSCCR cohort and MECONIUM cohort (parents and other family members), which are combined and shown by IBD group.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

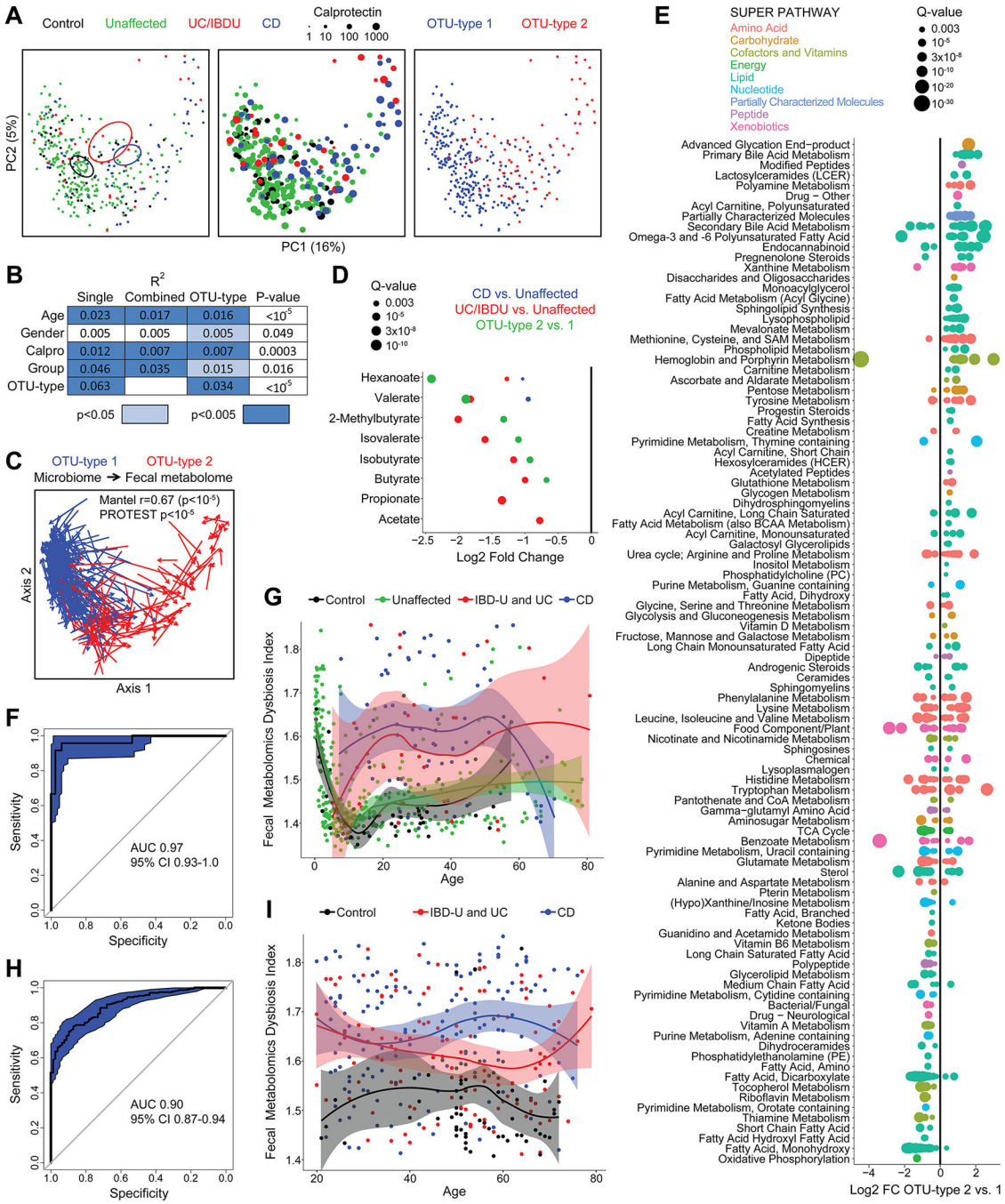


Figure 3. Dysbiosis is associated with reproducible shifts in the fecal metabolome.

(**A**) PCoA plots representing differences in fecal metabolomics profiles measured by Euclidean distance. Subjects under 3 years of age were excluded. (**B**) Effect size (marginal R^2) of factors contributing to variation in the fecal metabolome. (**C**) Microbiome and fecal metabolome PCoA plots were superimposed by Procrustes. Significance was determined by the Mantel test and PROTEST. Each arrow connects the microbiome to the fecal metabolome of the same sample. (**D**) Differentially abundant short chain fatty acids by OTU-type and IBD status adjusting for sex, age, and family. Dot size is proportional to

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

significance. (**E**) Differentially abundant fecal metabolites by OTU-type. Each dot represents one metabolite, organized into metabolic pathways. (**F**) A random forest classifier to predict OTU-type from fecal metabolites was constructed using a training subset of RTP. Classifier area under the ROC curve (AUC) with confidence interval calculated by bootstrapping is shown for samples in the test subset. (**G**) Fecal metabolite dysbiosis index is shown by age and IBD group. (**H**) Performance of the fecal metabolite OTU-type classifier trained on the RTP cohort for predicting OTU-types in the MSCCR cohort. (**I**) Fecal metabolite dysbiosis index is shown for the MSCCR cohort, divided by IBD group.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

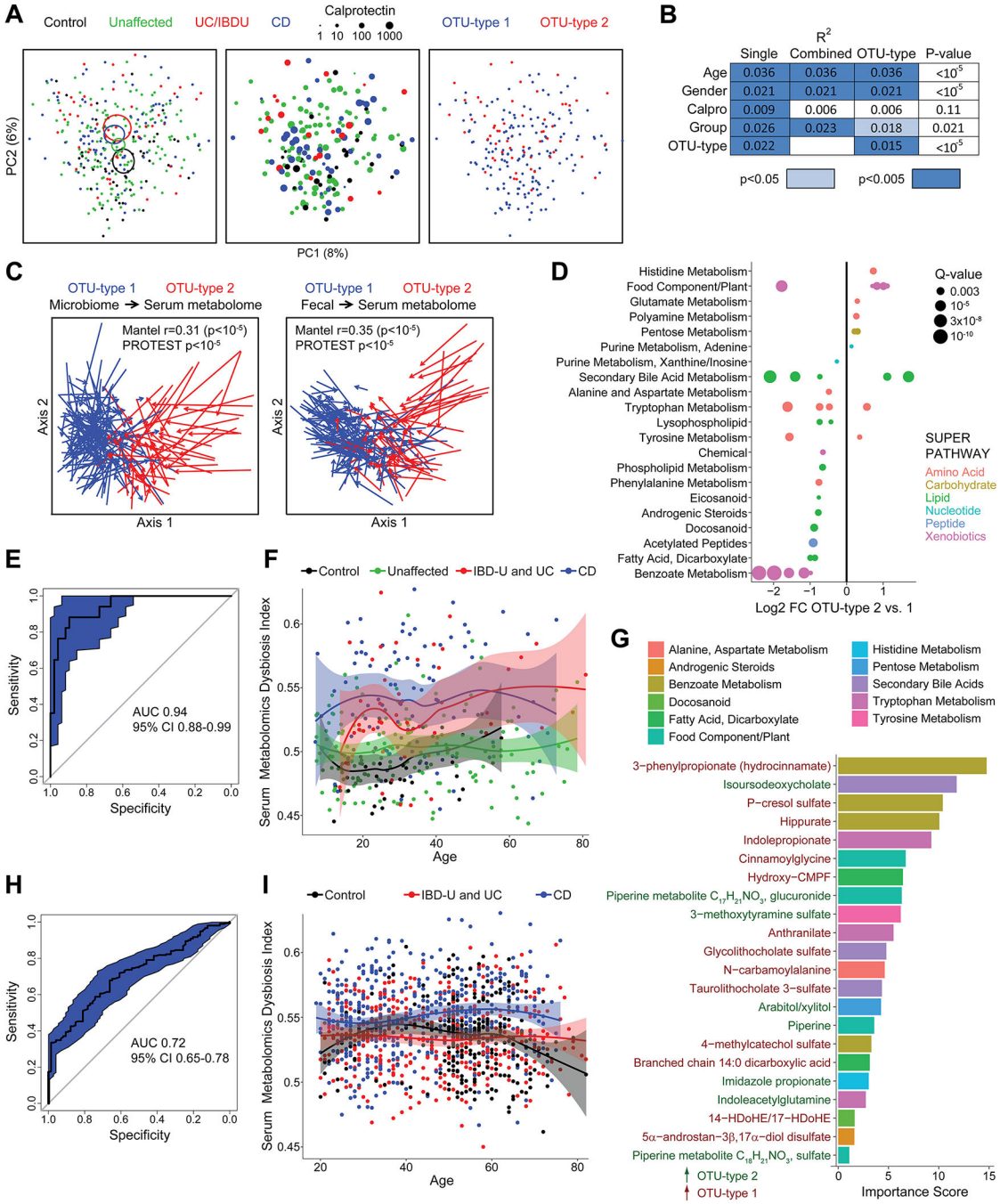


Figure 4. Serum metabolites reflect the gut microbiome and can indicate dysbiosis.

(**A**) PCoA plots of the serum metabolome, with each dot representing one subject colored by IBD group or OTU-type. (**B**) Effect size (R^2) of factors contributing to variation in the serum metabolome. (**C**) PCoA of the serum metabolome was superimposed onto microbiome and fecal metabolome PCoA plots by Procrustes. (**D**) Differentially abundant serum metabolites by OTU-type. (**E**) A random forest classifier to predict OTU-type from serum metabolites was constructed using a training subset of RTP. Classifier performance is shown for the test subset. (**F**) Serum metabolite dysbiosis index is shown by age and IBD group. (**G**)

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Importance scores for serum metabolites included in the OTU-type classifier. Metabolite name is colored by the direction of its association with OTU-type. Bars are colored by metabolic pathway. (**H**) Performance of the serum metabolite OTU-type classifier for predicting OTU-types in the MSCCR cohort. (**I**) Serum metabolite dysbiosis index is shown for the MSCCR cohort.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

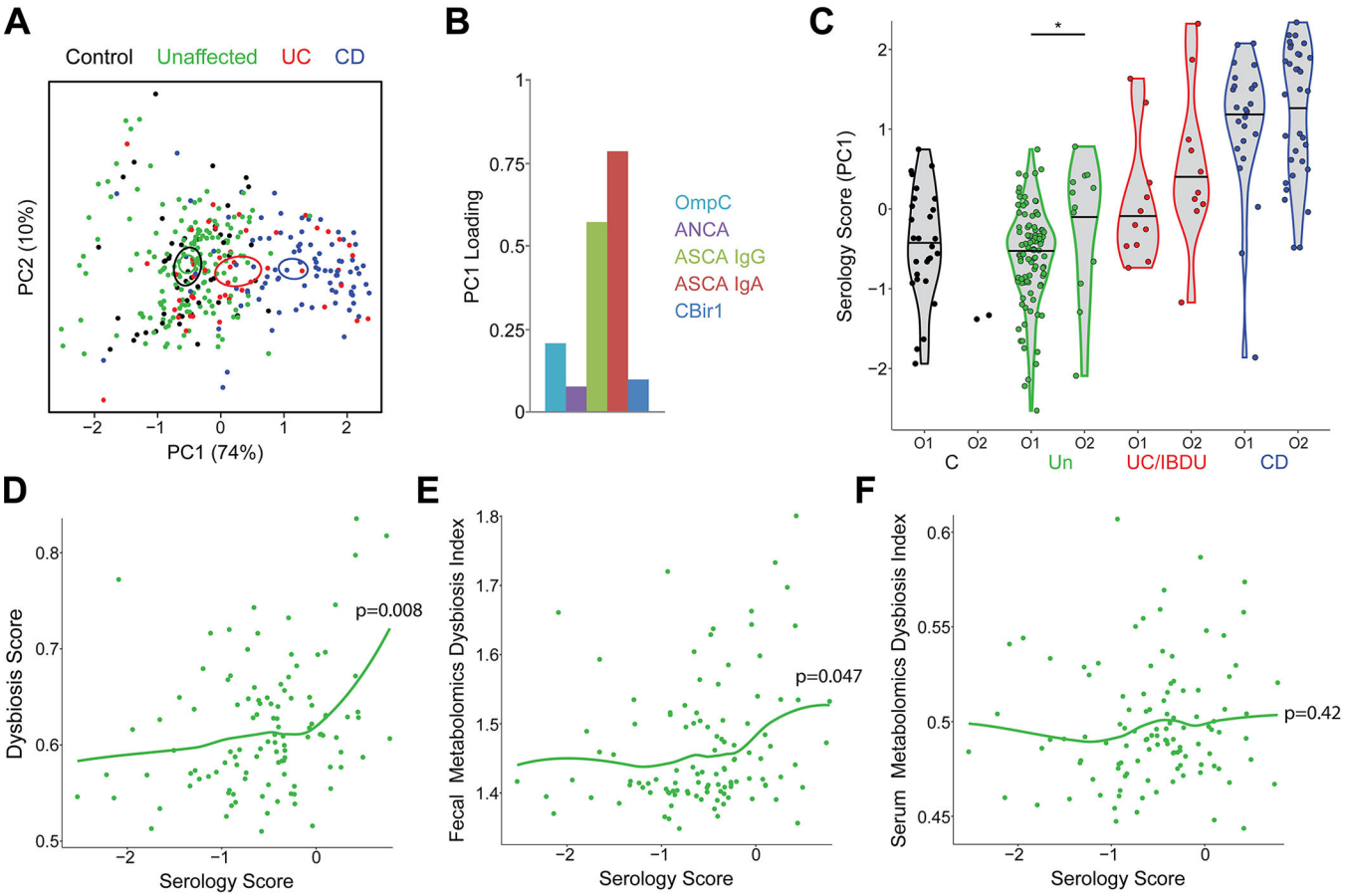


Figure 5. Dysbiosis is associated with increased IBD serologic markers in unaffected relatives.

(**A**) Principal components analysis (PCA) plot of IBD serologies in the RTP cohort, colored by IBD group. (**B**) Loading of the five IBD-associated antibodies on the horizontal axis of the PCA plot (PC1). (**C**) Serology score (equivalent to PC1) is shown for IBD groups divided by OTU-type. O1=OTU-type 1, O2= OTU-type 2. Significance was determined by Mann-Whitney U between OTU-types within each group. *p<0.05 (**D-F**) Association of serology score with (**D**) dysbiosis score, (**E**) fecal metabolomics dysbiosis index, and (**F**) serum metabolomics dysbiosis index in unaffected relatives. P-values were calculated using mixed effects models adjusting for age, sex, and family.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

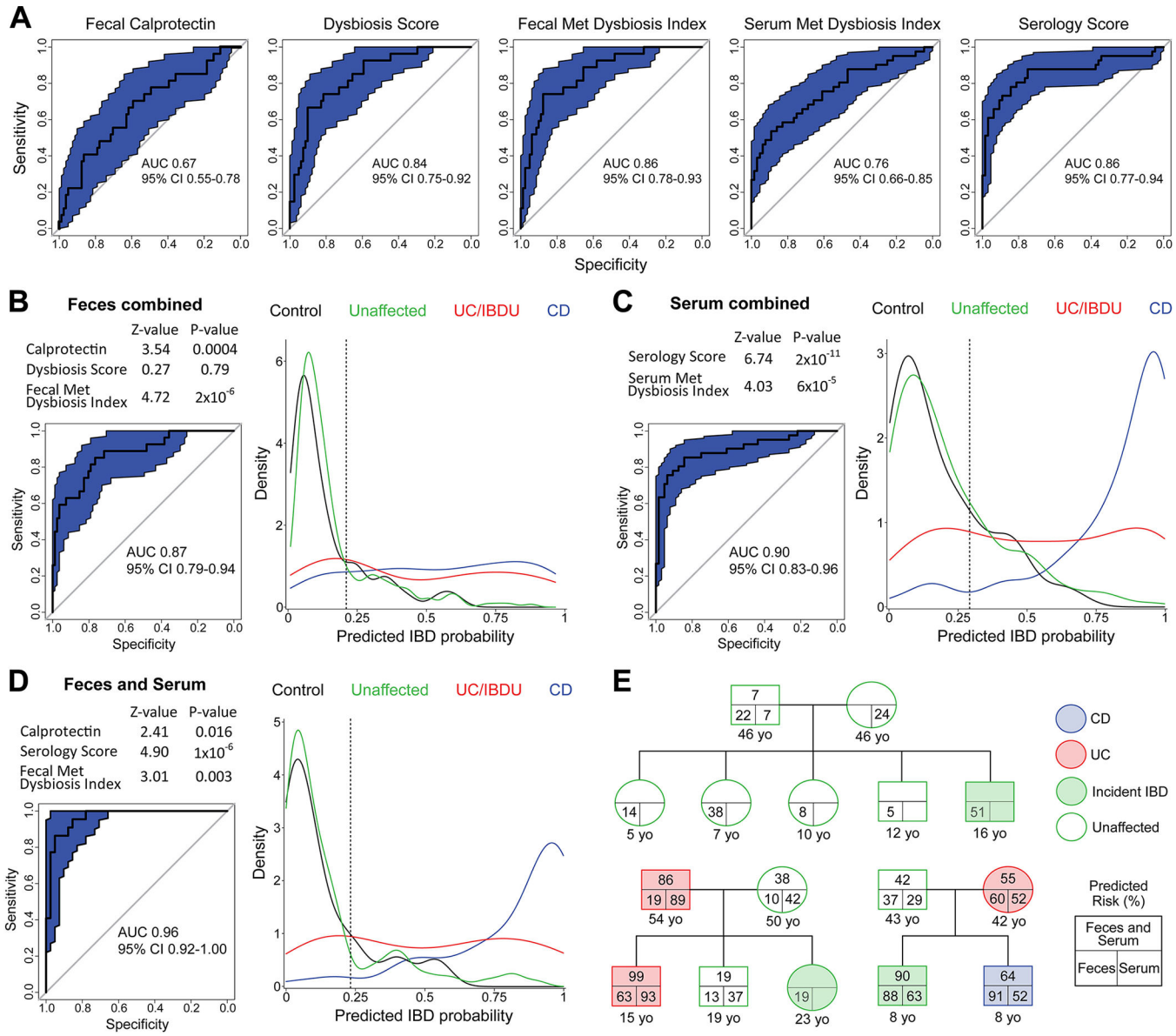


Figure 6. IBD risk models incorporating fecal and serum biomarkers of dysbiosis.

(**A**) ROC curves for univariate logistic regression models to differentiate IBD vs. unaffected family members. Models were created using a training subset and performance was estimated using a test subset. (**B-D**) Multivariate logistic regression models were built using combinations of (**B**) fecal and (**C**) serum biomarkers, or (**D**) both. For each model, the components are listed with their Z-score and p-value, the ROC curve is shown, and a histogram visualizes the distribution of predicted IBD probabilities. The dashed line demarcates low vs. high predicted IBD probability among unaffected individuals determined by two-component mixture models. (**E**) Pedigree charts for three unaffected family members who later developed IBD (incident IBD cases). IBD risk as estimated by multivariate logistic regression models (feces, serum, and both) is shown.

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 1.

Cohort Demographics and Clinical Features

	CD	UC/IBD-U	Unaffected	Control
N	64	31 (10 UC)	292	51
Number of families	37	18	45	10
Sex (% female)	53%	55%	52%	51%
Age	33.4 ± 14.4	36.3 ± 19.6	19.1 ± 20.3	20.8 ± 14.8
BMI	24.3 ± 5.0	24.7 ± 4.9	22.2 ± 6.6	24.9 ± 4.3
Calprotectin	243 (38–1572)	174 (35–873)	82 (21–327)	73 (13–411)
Age at Diagnosis	19.3 ± 10.2	25.6 ± 17.6		
Location/Extent	L1 39%, L2 12%, L3 47%, L4 2%	E1 14%, E2 57%, E3 14%, E4 29%		
Perianal	25%			
CD Behavior	B1 61%, B2 20%, B3 20%			
Surgery				
Colectomy	3%	10%		
Bowel Resection	20%	0%		
Medications				
5-ASA	77%	90%		
Immunomodulators	59%	40%		
Anti-TNF	61%	60%		
Anti-integrin	14%	10%		
Steroids	70%	78%		

Values shown as % or mean +/− standard deviation (geometric mean for fecal calprotectin). BMI is provided only for adults. Location/extent refers to the Paris classification for UC and the Montreal classification for CD.